

Andrea D'Angelo
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Short Bio

Andrea D'Angelo is a Danish Data Science Academy (DDSA) Post-doctoral researcher at Aarhus University, working on *Robust Graph Unlearning*. He was awarded a Ph.d. cum laude and Doctor Europaeus distinction at University of L'Aquila in 2026. His research focuses on Trustworthy Artificial Intelligence and Machine Unlearning. He has authored more than 15 papers published in leading venues (including IJCAI, CIKM, JSS, and IP&M), organized international workshops and delivered a tutorial at ECML-PKDD 2025, contributed to major conferences by serving as Proceedings Chair for CIKM 2024 and WSDM 2026, and delivered invited talks at several leading institutions, including the Technical University of Munich (TUM). He also holds the Big Data course at LUISS University of Rome, supervised five master's theses, and served on the program committees of major conferences.



Research Experience

- 2026- **DDSA Postdoc**, Datalogi, Aarhus University. Project: *Robust Graph Unlearning*. Funded by the Danish Data Science Academy.
- 2026 **Postdoctoral Researcher**, Datalogi, Aarhus University, (Host: Prof. Davide Mottin).
- 2026- Research Scholarship, University of L'Aquila, AI medical recommendation model (SIMPLIFY project) with Appendo.
- 2022-26 Ph.D. (XXXVIII cycle) Student, University of L'Aquila, (Advisors: Prof. Giovanni Stilo, Prof. Antiniscia Di Marco).
- 2025 6-month Visiting Ph.D. Student, Datalogi, Aarhus University, (Advisor: Prof. Davide Mottin). Worked on Graph Unlearning and Differential Privacy.
- 2021-22 **9 months Research Scholarship**, Data Analysis and Management for the "Territori Aperti" project, funded by CIGL Cisl UIL, University of L'Aquila. Worked on new techniques for LLM-based Dense Retrieval.

Education

- 2022-26 **Ph.D.**, Computer Science, University of L'Aquila, cum Laude, Doctor Europaeus. Thesis: "*Trustworthy AI Through Principled Advances in Machine Unlearning, Algorithmic Bias Mitigation, and Explainability*"
- 2020-22 **M.Sc.**, Computer Science (LM-18), University of L'Aquila, 110/110 cum Laude, 22/10/2022. Thesis: "*Clouds of vectors for density-based Information Retrieval*".

2016-19 **B.Sc.** Computer Science (L-31), University of L'Aquila, 110/110 cum Laude, 21/10/2019.
Thesis: "*Application of standard compilation techniques to non-linear languages*".

Honours & Awards

2026 Awarded **DDSA Postdoc Grant** (Danish Data Science Academy) for the project titled *Robust Graph Unlearning* (1,365,000 DKK / ~€183,000).
2025 Awarded **ERASMUS** Traineeship Grant for European mobility.
2024 Awarded CIKM (ACM International Conference on Information and Knowledge Management) 2024 **Distinguished Reviewer Award**.
2024 Awarded EUGAIN **Virtual Mobility Grant**, EUGAIN COST Action E-COST-GRANT-CA19122-04419a60.
2024 Awarded **Mobility Grant** from KDD 2024 (ACM SIGKDD Conference on Knowledge Discovery and Data Mining).
2024 Awarded **Best Short Paper** Award at ICPE 2024 (International Conference on Performance Engineering).
2022 Awarded publicly funded Ph.D. Scholarship at University of L'Aquila.

Teaching

LUISS University of Rome

2026 Lecturer, Big data management: data mining and analysis for management decisions, 21 hours

University of L'Aquila

2024 Co-Creator of the MOC course on "Reinforcement Learning" of the SoBigData Academy ¹.
2023, 2024 Assistance During Lectures, Deep Neural Networks (Computer Science Msc, LM-18), 8 hours.
2023, 2024 Assistance During Lectures, Artificial Intelligence (Computer Science Msc, LM-18), 5 hours.

Research Software



ERASURE is an open-source Python framework designed to make Machine Unlearning research reproducible and comparable. It provides a configuration driven interface for running and benchmarking forgetting algorithms across multiple models, datasets, and evaluation metrics. Published as a resource paper at **CIKM 2025** (Rank A) and demonstrated at **IJCAI 2025** (Rank A*).

¹<https://sobigdata.unipi.it/course/view.php?id=17>

Supervision

Scholars

- 2025 Francesca Ciccarelli, University of L'Aquila, Ph.D. Student, on the topics of Machine Unlearning applied to LLMs.
- 2025 Manuel Antonio Vilas Valiente, University of L'Aquila, Visiting Scholar, on the topics of Machine Unlearning applied to Time Series.
- 2024, 2025 Tutor for Pinkamp, a STEM Outreach program.

Theses

- 2026 Susanna Peretti, University of Urbino, Computer Science Master's Degree, "*Machine Unlearning applied to Time Series and Human Activity Recognition*"
- 2025 Fabio D'Andreamatteo, University of L'Aquila, Data Science Master's Degree, "*HybridKAN: Leveraging Multi-Sized Sub-MLPs for Enhanced Performances*".
- 2025 Samuele Ramassone, University of L'Aquila, Computer Science Master's Degree, "*Beyond the Systematic Evaluation of Large Language Models: From Theory to Practical Analysis of Results*".
- 2025 Gabriele Tagliente, University of L'Aquila, Computer Science Master's Degree, "*From ERASURE to Evaluation of Machine Unlearning Techniques*".
- 2025 Matteo Iezzi, University of L'Aquila, Data Science Master's Degree, "*Recent Advances in Chain-of-Thought Prompting: Current Trends and Automation Approaches*".

Service

Organization

- 2026 **Co-organizer** of Parallel Session "Knowledge Graphs: Connecting Theory, Algorithms, and Applications", D3A 3.0 (Danish Digitalization, Data Science and AI), Nyborg Strand, Denmark.
- 2026 **Co-organizer** of Workshop on Innovations, Privacy-preservation, and Evaluations of machine Unlearning Techniques (WIPE-OUT 2), ECML-PKDD (European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases) 2026.
- 2025 **Co-organizer** of Workshop on Web & Graphs, Responsible Intelligence, and Social Media (WEB&GRAPH 2026), WSDM (The 19th ACM International Conference on Web Search and Data Mining) 2026, Boise, Idaho, USA.
- 2025 **Proceedings Chair** of WSDM (ACM International Conference on Web Search and Data Mining) 2026.
- 2025 **Co-organizer** of Workshop on Innovations, Privacy-preservation, and Evaluations of machine Unlearning Techniques (WIPE-OUT 2025), ECML-PKDD (European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases) 2025, Porto.

- 2025 **Program Committee Chair** of Workshop on Innovations, Privacy-preservation, and Evaluations of machine Unlearning Techniques (WIPE-OUT 2025), ECML-PKDD (European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases) 2025, Porto.
- 2025 **Co-organizer** of Tutorial: Unraveling Graph Counterfactual Explainability: from Theoretical Foundations to Technical Mastery, ECML-PKDD (European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases) 2025, Porto.
- 2024 **Proceedings Chair** of CIKM (ACM International Conference on Information and Knowledge Management) 2024.
- 2024 **Co-organizer** of "Territori Aperti: dai risultati alle sfide future" event.

Member of the Program Committee

- 2026 Springer Artificial Intelligence Review Journal
- 2026 Automated Software Engineering Journal
- 2025 ACM Web Conference (formerly known as the International World Wide Web Conference)
- 2025, 2026 KDD (ACM SIGKDD Conference on Knowledge Discovery and Data Mining) Datasets & Benchmarks.
- 2024, 2025, 2026 CIKM (ACM International Conference on Information and Knowledge Management). **Distinguished Reviewer** in 2024.
- 2025, 2026 ECML-PKDD (European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases).
- 2025 ICPE Data Challenge (International Conference on Performance Engineering).
- 2024, 2026 LL4MSE Workshop @ STAF (Software Technologies: Applications and Foundations).
- 2024 DELTA Workshop @ KDD 2024 (ACM SIGKDD Conference on Knowledge Discovery and Data Mining).
- 2024 MIDAS Workshop @ ECML-PKDD 2024 (European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases).
- 2023 BIAS Workshop @ ECIR 2023 (European Conference on Information Retrieval).

Presentations

Invited Talks

- IT1. **D'Angelo, Andrea.** *Opportunities and Privacy Risks with Artificial Intelligence* Istituto Tecnico Marconi Penne. 2026.
- IT2. **D'Angelo, Andrea.** *Perspectives on Machine Unlearning and Privacy-preserving Machine Learning*, Technical University of Dresden. 2026.
- IT3. **D'Angelo, Andrea.** *Forgetting Right: Towards Reproducible Benchmarks in Machine Unlearning with ERASURE* WIPEOUT Workshop @ ECML PKDD 2025 (European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases), Porto. 2025.
- IT4. **D'Angelo, Andrea.** *Machine Unlearning: Safeguarding Privacy and Security in AI Models* Technical University of Munich. July 2024.

- IT5. **D’Angelo, Andrea.** *Privacy Implications in the Training and Use of Large Language Models* Territori Aperti, Napoli. 2024.
- IT6. **D’Angelo, Andrea.** *Uncovering Gender Bias in Academia* AIGAP 2023 (Algorithmic Biases in Artificial Intelligence from Interdisciplinary Perspectives), L’Aquila. 2023.

Talks

- T1. **D’Angelo, Andrea.** *HybridKAN: Leveraging Multi-Sized Sub-MLPs for Enhanced Performance* ITADATA (Italian Conference on Big Data and Data Science) 2025, Turin. 2025.
- T2. **D’Angelo, Andrea.** *A Novel Relevance Score for Unsupervised Retrieval with Large Language Models* KDD 2024 (ACM SIGKDD Conference on Knowledge Discovery and Data Mining), Barcelona. 2024.
- T3. **D’Angelo, Andrea.** *Echocardiographic Epicardial Adipose Tissue Quantification: Challenges and Insights*; Percom (International Conference on Pervasive Computing and Communications) 2024, Biarritz. 2024.
- T4. **D’Angelo, Andrea.** *Data-Driven Analysis of Gender Fairness in the Software Engineering Academic Landscape* ECSA 2023 (European Conference on Software Architecture), Istanbul. 2023.

Posters

- P1. **D’Angelo, Andrea.** *A novel Relevance Score for Unsupervised Retrieval with Large Language Models* KDD (ACM SIGKDD Conference on Knowledge Discovery and Data Mining) 2024, Barcelona, Spain.
- P2. Ciccarelli, F., **D’Angelo, Andrea** & Stilo, G. *Towards a Novel Visual Evaluation of Algorithmic Bias: Insights on the Italian Academic System* KDD (ACM SIGKDD Conference on Knowledge Discovery and Data Mining), 2024.

Tutorials

- D1. Prenkaj, B., Zaradoukas, E., Stilo, G., **D’Angelo, Andrea** & Prado-Romero, M. *Unraveling Graph Counterfactual Explainability: from Theoretical Foundations to Technical Mastery* ECML-PKDD 2025, Porto, Portugal. Sept. 2025.

Publications

G Google Scholar: <https://scholar.google.com/citations?user=9WIwwAAAAJ>

Book Chapters

- B1. Takaoka, A. J. W., Alboaie, L., Cutrupi, C. M., **Andrea D’Angelo**, Marco, A. D. & Hillston, J. *Best Practices and Lessons from Academia-Industry Collaboration Initiatives* Mar. 2025.

Journal Articles

Reported journal ranks are from Scimago Journal rank (SJR).

- J1. D'Angelo, A., Savelli, C., Giobergia, F., Baralis, E. & Stilo, G. On the Evaluation of Machine Unlearning Methods: A Multi-Domain Classification Benchmark. *Machine Learning [Q1]* (2026).
- J2. D'Angelo, A., Gullo, F. & Stilo, G. The Forget-Set Identification Problem. *Machine Learning [Q1]* 114. <https://doi.org/10.1007/s10994-025-06897-9> (2025).
- J3. **Andrea D'Angelo**, d'Aloisio, G., Marzi, F., Di Marco, A. & Stilo, G. Uncovering gender gap in academia: A comprehensive analysis within the software engineering community. *Journal of Systems and Software [Q1]* 217, 112162. ISSN: 0164-1212. <https://www.sciencedirect.com/science/article/pii/S0164121224002073> (2024).
- J4. Bianchi, A., Zelli, V., **D'Angelo, Andrea**, Di Matteo, A., Scoccia, G., Cannita, K., Dimas, A. S., Glentis, S., Zazzeroni, F., Alesse, E., Di Marco, A. & Tessitore, A. A method to comprehensively identify germline SNVs, INDELs and CNVs from whole exome sequencing data of BRCA1/2 negative breast cancer patients. *NAR Genomics and Bioinformatics [Q1]* 6, lqae033. ISSN: 2631-9268. eprint: <https://academic.oup.com/nargab/article-pdf/6/2/lqae033/57258946/lqae033.pdf>. <https://doi.org/10.1093/nargab/lqae033> (Apr. 2024).
- J5. d'Aloisio, G., **Andrea D'Angelo**, Di Marco, A. & Stilo, G. Debiasser for Multiple Variables to enhance fairness in classification tasks. *[Information Processing & Management] [Q1]* 60, 103226. ISSN: 0306-4573 (2023).
- J6. Della Penna, G., Orefice, S. & **D'Angelo, A.** Exploiting spatial relations for grammar-based specification of multidimensional languages. *Knowledge and Information Systems [Q1]* 65, 3995–4020 (2023).

Peer-reviewed

Conference Papers. Reported conference ranks are from CORE.

- C1. D'Andreanmatteo, F., **D'Angelo, Andrea** & Stilo, G. *HybridKAN: leveraging Multi-Sized Sub-MLPs for Enhanced Performances in ITADATA (Italian Conference on Big Data and Data Science) 2025* (Torino, Italy, 2025).
- C2. **D'Angelo, A.**, Savelli, C., Tagliente, G., Giobergia, F., Baralis, E. & Stilo, G. *How to Make Reproducible Research in Machine Unlearning with ERASURE in Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence, IJCAI-25 [A*] Demo Track* (International Joint Conferences on Artificial Intelligence Organization, Aug. 2025), 11025–11029. <https://doi.org/10.24963/ijcai.2025/1255>.
- C3. **D'Angelo, Andrea**, Savelli, C., Tagliente, G., Giobergia, F., Baralis, E. & Stilo, G. *ERASURE: a Modular and Extensible Framework for Machine Unlearning in Proceedings of the 34th ACM International Conference on Information and Knowledge Management (CIKM 2025 [A])* (Association for Computing Machinery, Seoul, Korea, 2025).
- C4. Di Sipio, C., **D'Angelo, Andrea**, Rubei, R. & Politowski, C. *On the Need for Reproducibility Guidelines for Open-Source Games: A itch.io Case Study in 2025 IEEE Conference on Games (CoG) [C]* (2025), 1–8.

- C5. d'Aloisio, G., **D'Angelo, Andrea**, Marzi, F., Di Marco, D., Stilo, G. & Di Marco, A. *Data-Driven Analysis of Gender Fairness in the Software Engineering Academic Landscape in Software Architecture. ECSA 2023 Tracks, Workshops, and Doctoral Symposium [A]* (eds Tekinerdoğan, B., Spalazzese, R., Sözer, H., Bonfanti, S. & Weyns, D.) (Springer Nature Switzerland, Cham, 2024), 89–103. ISBN: 978-3-031-66326-0.
- C6. **D'Angelo, Andrea** & d'Aloisio, G. *Grammar-Based Anomaly Detection of Microservice Systems Execution Traces in Companion of the 15th ACM/SPEC International Conference on Performance Engineering [B]* (Association for Computing Machinery, London, United Kingdom, 2024), 77–81. ISBN: 9798400704451. <https://doi.org/10.1145/3629527.3651844>. Best Short Paper Award (ICPE 2024).
- C7. **D'Angelo, Andrea**, Di Sipio, C., Politowski, C. & Rubei, R. *PlayMyData: a curated dataset of multi-platform video games in Proceedings of the 21st International Conference on Mining Software Repositories [A]* (Association for Computing Machinery, Lisbon, Portugal, 2024), 525–529. ISBN: 9798400705878. <https://doi.org/10.1145/3643991.3644869>.
- C8. Bianchi, A., d'Aloisio, G., **Andrea D'Angelo**, Marco, A. D., Matteo, A. D., Leone, J., Scoccia, G., Stilo, G. & Traini, L. *DIORAMA: Digital twIn fOR sustAinable territorial MAnagement in Proceedings of itaDATA (Italian Conference on Big Data and Data Science)* (Sept. 2022), 144–155.
- C9. d'Aloisio, G., Stilo, G., Di Marco, A. & **D'Angelo, Andrea**. *Enhancing Fairness in Classification Tasks with Multiple Variables: A Data- and Model-Agnostic Approach in Advances in Bias and Fairness in Information Retrieval* (eds Boratto, L., Faralli, S., Marras, M. & Stilo, G.) (Springer International Publishing, Cham, 2022), 117–129. ISBN: 978-3-031-09316-6.

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